

97.2%

HW 3 Austin Tartaglia

1a) $P = 100 \text{ lbf/in}^2$ $u = 500 \text{ Btu/lb}$

$T = 328^\circ \text{F}$ $v = 1.12 \text{ ft}^3/\text{lb}$ $h = 521 \text{ Btu/lb}$

1b) Ammonia @ 40°F

$P = 73.3 \text{ lbf/in}^2$

$v = v_f + x(v_g - v_f)$
 $= 0.0233 + .5(3.97 - 0.0233)$

$v = 1.9977 \text{ ft}^3/\text{lb}$

$u = u_f + x(u_g - u_f)$

$= 86.3 + .5(869 - 86.3)$

$u = 327.65 \text{ Btu/lb}$

1c) Water @ 80 bar & $v = 0.04084 \text{ m}^3/\text{kg}$

$T = 480^\circ \text{C}$ $u = 3030 \text{ kJ/kg}$ $h = 3350 \text{ kJ/kg}$

1d) $T = -20^\circ \text{C}$ $h = 235.81 \text{ kJ/kg}$

$P \approx 1.33 \text{ bar}$ $v = 0.145 \text{ m}^3/\text{kg}$ $u = 216 \text{ kJ/kg}$

2a) $T_1 = 180$
 $T_2 = 100$

$P_1 = 1.0322 \text{ MPa}$ $v_1 = 0.19405 \text{ m}^3/\text{kg}$

$v_1 = v_2 = 0.19405 \text{ m}^3/\text{kg}$

@ 100°C $v_{f2} = 1.0435 \times 10^{-3}$
 $v_{g2} = 1.673 \text{ m}^3/\text{kg}$
 $v_{f2} < v_2 < v_{g2}$

$P_2 = 1.014 \text{ MPa}$

2b) $x_2 = \frac{v_2 - v_{f2}}{v_{g2} - v_{f2}}$

$= \frac{0.19405 - 1.0435 \times 10^{-3}}{1.673 - 1.0435 \times 10^{-3}} = x = 0.115$

3a) $P = 300 \text{ kPa} = 3 \text{ bar}$
 $V_1 = 2.064 \text{ m}^3$
 $\gamma_1 = 0.6058 \text{ m}^3/\text{kg}$

$T_1 = 133.6^\circ\text{C}$

3b) $V_2 = \frac{V_1}{m} = \frac{2.064 \text{ m}^3}{2 \text{ kg}} = 1.032 \text{ m}^3/\text{kg}$

$V_2 = 1.032 \text{ m}^3/\text{kg}$

$P_2 = 300 \text{ kPa}$

~~$1.032 \text{ m}^3/\text{kg}$~~ m^3/kg , not m^3/kg
 from table A-4

$T_2 = 400^\circ\text{C}$

3c) $w = P(V_2 - V_1)$

$w = (300 \frac{\text{KN}}{\text{m}^2})(2.064 - 1.032 \text{ m}^3)$

$w = 309.6 \text{ KN} \cdot \text{m}$

$w = 309.6 \text{ kJ}$

80%

4a) $P_1 = 10 \text{ bar}$ $P_2 = 2 \text{ bar}$ $m = 1 \text{ kg}$
 $T_1 = 60^\circ\text{C}$ $T_2 = 30^\circ\text{C}$

$V_1 = m \cdot \gamma_1$ $\gamma_1 @ 10 \text{ bar} + 60^\circ\text{C} = 0.02301 \frac{\text{m}^3}{\text{kg}}$

$V_1 = (1 \text{ kg})(0.02301 \frac{\text{m}^3}{\text{kg}}) = 0.02301 \text{ m}^3 = V_1$

4b) $U_1 = 268.35 \frac{\text{kJ}}{\text{kg}}$ $U_2 = 253.06 \frac{\text{kJ}}{\text{kg}}$

$U_1 = m \cdot U_1 = (1 \text{ kg})(268.35 \frac{\text{kJ}}{\text{kg}}) = 268.35 \text{ kJ}$

$U_2 = m \cdot U_2 = (1 \text{ kg})(253.06 \frac{\text{kJ}}{\text{kg}}) = 253.06 \text{ kJ}$

$\Delta U = U_2 - U_1 = 253.06 - 268.35 \text{ kJ} = -15.29 \text{ kJ}$

$\Delta U + \cancel{\overset{=0}{KE}} + \cancel{\overset{=0}{PE}} = Q - W$

$-15.29 \text{ kJ} = Q - 20 \text{ kJ}$

$Q = -15.29 + 20 \text{ kJ}$

$Q = 4.71 \text{ kJ}$

5a) 1.5 kg H₂O

$$T_1 = 70^\circ\text{C} \quad T_2 = 120^\circ\text{C}$$

$$P_1 @ 70^\circ\text{C} = 0.3119 \text{ bar} \times \frac{100 \text{ kPa}}{1 \text{ bar}} = 31.19 \text{ kPa}$$

$$5b) @ 120^\circ\text{C} \quad v_g = 0.8919 \frac{\text{m}^3}{\text{kg}} \\ u_g = 5259.3 \frac{\text{kJ}}{\text{kg}}$$

$$@ 70^\circ\text{C} \quad v_{1g} = 5.042 \quad v_{1f} = 1.0228 \times 10^{-3} \\ u_{1f} = 292.25 \quad u_{1g} = 2469.6$$

$$u_1 = \frac{v_{1g} - v_{1f}}{v_{1g} - v_{1f}} (u_{1g} - u_{1f}) + u_{1f} = \frac{5.042 - 0.8919}{5.042 - 1.0228 \cdot 10^{-3}} (2469.6 - 292.25) + 292.25 \\ = \frac{4.1501}{5.0409} (2177.35) + 292.25 = 2084.83 \frac{\text{kJ}}{\text{kg}}$$

$$U_1 = m \cdot u_1 = (1.5 \text{ kg}) (2084.83 \frac{\text{kJ}}{\text{kg}}) = 3127.245 \text{ kJ}$$

$$U_2 = m \cdot u_2 = (1.5 \text{ kg}) (5259.3 \frac{\text{kJ}}{\text{kg}}) = 7888.95 \text{ kJ}$$

$$\Delta U = U_2 - U_1 = 7888.95 - 3127.245 = 4761.7 \text{ kJ}$$

$$\Delta U + \cancel{KE} + \cancel{PE} = Q - W$$

$$4761.705 \text{ kJ} = Q$$

80%

6a) R₂₂ @ 10 bar

$$V = m \cdot v$$

$$m = 25 \text{ kg} \quad x = 60\% \text{ or } 0.6$$

$$v = v_f + x(v_g - v_f) = 0.8352 \times 10^{-3} + 0.6(0.0236 - 0.8352 \times 10^{-3}) \\ = 0.8352 \times 10^{-3} + 0.01366 = 0.0145 \frac{\text{m}^3}{\text{kg}}$$

$$m = \frac{m_f}{1-x} = \frac{25}{1-0.6} = 62.5 \text{ kg}$$

$$V = m \cdot v = (62.5 \text{ kg}) (0.0145 \frac{\text{m}^3}{\text{kg}}) = 0.906 \text{ m}^3 = V$$

$$c) V_{gM} = V_g(M - M_c) = 0.0236(62.5 - 23) \\ = \frac{0.885 \text{ m}^3}{0.906 \text{ m}^3} = 0.977 \times 100 = \boxed{97.7\%}$$

$$7a) 0.2 \text{ g He} \\ T_1 = 27^\circ\text{C} \quad P_1 = 1.3 \text{ bar} \\ T_2 = 17^\circ\text{C} \quad P_2 = 0.8 \text{ bar}$$

$$R = \frac{\bar{R}}{M_{\text{He}}} = \frac{8.314 \frac{\text{J}}{\text{mol} \cdot \text{K}}}{4.003 \text{ g/mol}} = \boxed{2.077 \frac{\text{J}}{\text{g} \cdot \text{K}} = R}$$

$$7b) PV = nRT \quad V = \frac{nRT}{P} = \frac{(0.2) (2.077) (27 + 273)}{1.3 \times 10^5 \text{ Pa}} = \frac{124.62 \text{ J}}{1.3 \times 10^5 \text{ Pa}} = 9.586 \times 10^{-4} \text{ m}^3 \times \frac{100^3 \text{ cm}^3}{1^3 \text{ m}^3} \\ \boxed{V_1 = 958.6 \text{ cm}^3}$$

$$7b) V = \frac{(0.2) (2.077) (17 + 273)}{0.8 \times 10^5} = \frac{126.466}{80000} = 0.0015808 \text{ m}^3 \times \frac{100^3 \text{ cm}^3}{1^3 \text{ m}^3} = \boxed{158.08 \text{ cm}^3 = V_2}$$

$$8a) T_1 = 25^\circ\text{C} \quad P_1 = 2.75 \text{ bar} \quad V_1 = 0.05 \text{ m}^3 \\ T_2 = 257^\circ\text{C} \quad Q = -2.75 \text{ kJ}$$

$$R = \frac{8.314}{44.01} = 0.1889 \frac{\text{J}}{\text{g} \cdot \text{K}} \times 1000 = 188.9 \frac{\text{J}}{\text{kg} \cdot \text{K}}$$

$$PV = nRT \quad m = \frac{PV}{RT} = \frac{(2.75 \times 10^5)(0.05)}{(188.9)(25 + 273)} = \frac{13750}{56292.2} = \boxed{0.244 \text{ kg} = m_{\text{CO}_2}}$$

$$8b) \Delta U + \cancel{\Delta E} + \cancel{\Delta E} = Q - W \\ \Delta U = -2.75 \text{ kJ} - W \\ \Delta U = U_2 - U_1$$

$$U_1 = 6585 \quad U_2 = 14622$$

$$\Delta U = m \left(\frac{U_2 - U_1}{M_{\text{CO}_2}} \right) = 0.244 \left(\frac{14622 - 6585}{44.01} \right) = 42.89 \text{ kJ}$$

$$\frac{42.89 \text{ kJ} + 2.75 \text{ kJ}}{-1} = \boxed{W = -45.64 \text{ kJ}}$$